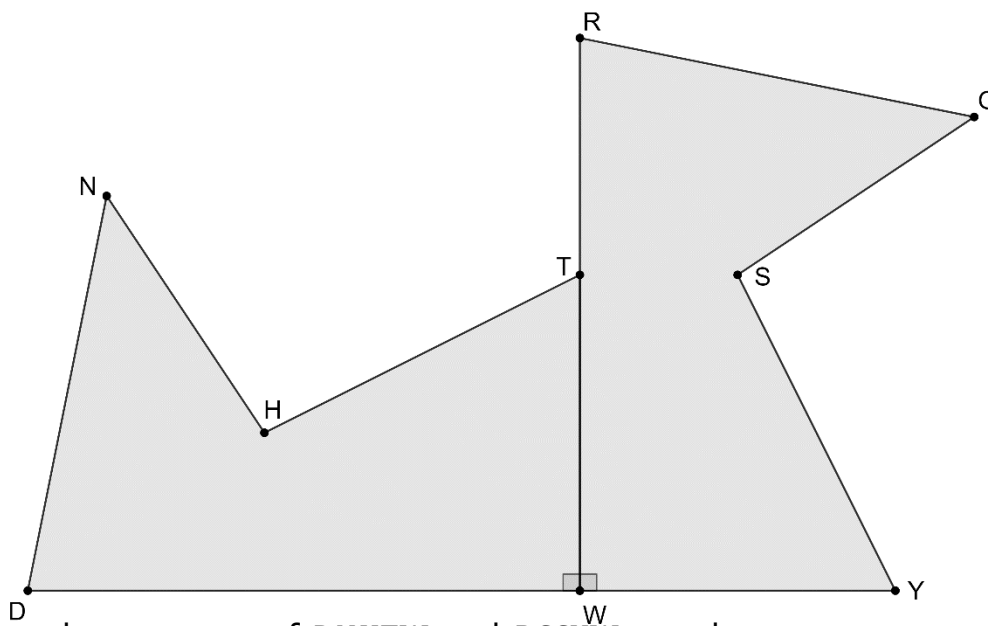


Geometry
Unit 5
Lesson 9 Practice

Name _____
Date _____
Period _____

Congruent polygons $DNHTW$ and $RGSYW$ are shown, where $\angle DWT$ and $\angle RWY$ are right angles.



Some of the angle measures of $DNHTW$ and $RGSYW$ are shown.

$$m\angle HTW = (6x - 15)^\circ$$

$$m\angle NHT = (12k + 14)^\circ$$

$$m\angle DNH = (3x + 6)^\circ$$

$$m\angle WRG = (3k + 13)^\circ$$

$$m\angle WYS = (4x + 11)^\circ$$

$$m\angle YWR = (4k + 6)^\circ$$

1. Determine the value of k .

2. Determine the value of x .

Determine each angle measure.

3. $m\angle GSY = \underline{\hspace{2cm}}$

4. $m\angle RGS = \underline{\hspace{2cm}}$

5. $m\angle WDN = \underline{\hspace{2cm}}$

Some of the measures of the side lengths of $DNHTW$ and $RGSYW$ are shown.

$$ND = 5h + 33$$

$$HT = 7h - 33$$

$$RW = 9h - 6$$

$$GS = 2h + 12$$

$$TW = 5h$$

$$DW + WY = 330$$

6. Determine the value of h .

7. What is the length of $\overline{HN}$?

9. What is the length of $\overline{RG}$?

8. What is the length of $\overline{SY}$?

10. What is the length of $\overline{DY}$?